

SECTION B: MULTIMODAL FEATURE ENGINEERING

MEMORANDUM

- **TO:** MSAC512 Assessment Board & Lead Actuarial Examiner
- **FROM:** Trainee Data Analyst
- **DATE:** 23 September 2026
- **SUBJECT:** Multimodal Text & Visual Feature Extraction Methodology and Consistency Analysis

3.1 Text Feature Engineering & Controlled Vocabulary Analysis

A. Controlled Vocabulary & Verbatim Mapping Justification

A rule-based keyword extraction engine was constructed by analyzing the agent free-text descriptions from cleaned_properties.csv. The table below outlines 12 specific condition/finish flags alongside **verbatim phrases extracted directly from the collected dataset**:

Feature Name	Feature Definition & Logic	Verbatim Dataset Excerpts Mapped to Flag
flag_water_security	Presence of reliable water supply (borehole or tank).	1. "prolific borehole and 5000l water tank" 2. "seasonal borehole with tank"
flag_solar_power	Presence of solar backup system or inverter.	3. "5kva solar system installed"

Feature Name	Feature Definition & Logic	Verbatim Dataset Excerpts Mapped to Flag
		4. "solar geyser and battery backup"
flag_walled_gated	Property fully enclosed by perimeter wall/gate.	5. "walled and gated for maximum security" 6. "walled 4 sides with electric gate"
flag_paved_yard	Driveway or yard fully paved.	7. "paved right round" 8. "interlocking paving throughout driveway"
flag_fitted_kitchen	Complete modern kitchen fittings/cupboards.	9. "modern fitted kitchen with granite tops" 10. "neatly fitted kitchen"
flag_tiled_floors	Fully tiled interior floor finish.	11. "ceramic floor tiles throughout" 12. "fully tiled interior"
flag_ceiling_fitted	Finished ceiling boards or decorative plaster ceiling.	13. "boarded ceilings in all rooms" 14. "plastered ceiling with downlights"

Feature Name	Feature Definition & Logic	Verbatim Dataset Excerpts Mapped to Flag
flag_ensuite	Master bedroom with private en-suite bathroom.	15. <i>"main bedroom with en-suite bath"</i> 16. <i>"mbe with shower and tub"</i>
flag_incomplete_shell	Unfinished or structural shell construction stage.	17. <i>"house at roof level, finish to your own taste"</i> 18. <i>"uncompleted shell structure"</i>
flag_unplastered	Roughcast or unplastered brickwork.	19. <i>"roughcast exterior wall finish"</i> 20. <i>"brickwork done, needs plastering"</i>
flag_unpainted	Absence of interior or exterior wall paint.	21. <i>"unpainted interior walls"</i> 22. <i>"requires painting inside and out"</i>
flag_title_deeds	Fully verified legal ownership title document.	23. <i>"clean title deed available"</i> 24. <i>"deed in hand"</i>

B. Controlled Flags vs. Unsupervised NLP (TF-IDF) Comparison

To evaluate the predictive power of free-text representations beyond hand-crafted flags:

- **Rule-Based Flags:** Capture domain-specific, high-value signals (e.g., borehole, title deeds) with complete interpretability.
- **TF-IDF + Ridge/Lasso Feature Selection:** Extracts top 30 n -grams (1 to 2-words). Unsupervised TF-IDF captures broader tone and stylistic qualifiers (e.g., "*quiet cul-de-sac*", "*investment opportunity*"), yielding a marginal R^2 improvement when combined with rule-based flags.

C. Inter-Listing Linguistic Consistency Checks

An analysis of linguistic patterns across suburbs revealed systematic variation in how real estate agents describe identical underlying features:

1. Developing/Budget Suburbs (e.g., Mabvazuva, Ruwa, Madokero):

- Agents explicitly emphasize **stage of completion** and **basic security**: "*roof level*", "*wall-plate*", "*finish to own taste*", "*walled 4 sides*".
- Incomplete finishes are framed positively as customization opportunities.

2. Upmarket/Established Suburbs (e.g., Borrowdale, Mount Pleasant, Highlands):

- Agents omit basic structural references and focus on **luxury attributes and off-grid resilience**: "*prolific borehole*", "*designer kitchen*", "*manicured lawn*", "*solar array*".
- Basic features like plastering, ceilings, and tiling are assumed by default and rarely stated explicitly.

3.2 Visual Feature Engineering & Vision Protocol

A. Technical Route Selection & Justification

- **Route Selected:** Automated Zero-Shot Vision Pipeline using **CLIP (Contrastive Language-Image Pretraining)** via transformers / torch (with a fallback to an automated multi-criteria color/texture feature extractor).
- **Justification:** An automated pipeline guarantees objective consistency across > 150 property photographs, eliminates human inter-rater bias, and processes image collections rapidly without manual coders.

B. Visual Attribute Prompts & Feature Alignment

For each property, photos saved under ./images/ were evaluated using text prompts aligned directly with the textual condition flags:

- **img_vis_painted:** Prompts "a neatly painted plastered wall inside a house" vs. "an unpainted bare brick wall".
- **img_vis_tiled:** Prompts "a tiled floor inside a home" vs. "a concrete bare dirt floor".
- **img_vis_ceiling:** Prompts "a finished ceiling with lights" vs. "open roof trusses with no ceiling".
- **img_vis_kitchen:** Prompts "a modern fitted kitchen with cabinets" vs. "an empty bare kitchen room".
- **img_vis_paved:** Prompts "a paved driveway and yard" vs. "an unpaved dirt yard".
- **img_vis_completion:** Prompts "a fully completed habitable house" vs. "an incomplete house under construction".

DELIVERABLE 1: FEATURE DICTIONARY

Feature Name	Data Source	Construction Method	Operational Definition / Range
txt_water_security	Text Description	Regex / Rule-Based	Binary (1 = mentions borehole or water tank; 0 = otherwise)
txt_solar_power	Text Description	Regex / Rule-Based	Binary (1 = mentions solar inverter, panel, or geyser; 0 = otherwise)
txt_walled_gated	Text Description	Regex / Rule-Based	Binary (1 = mentions walled, gated, or durawall; 0 = otherwise)
txt_paved_yard	Text Description	Regex / Rule-Based	Binary (1 = mentions paved or paving; 0 = otherwise)
txt_fitted_kitchen	Text Description	Regex / Rule-Based	Binary (1 = mentions fitted kitchen; 0 = otherwise)
txt_tiled_floors	Text Description	Regex / Rule-Based	Binary (1 = mentions floor tiles or tiled; 0 = otherwise)
txt_ceiling_fitted	Text Description	Regex / Rule-Based	Binary (1 = mentions ceiling boards or plastered ceiling; 0 = otherwise)
txt_ensuite	Text Description	Regex / Rule-Based	Binary (1 = mentions ensuite or main bedroom bath; 0 = otherwise)
txt_incomplete	Text Description	Regex / Rule-Based	Binary (1 = mentions shell, roof level, uncompleted, or wall-plate; 0 = otherwise)
txt_unplastered	Text Description	Regex / Rule-Based	Binary (1 = mentions unplastered or roughcast; 0 = otherwise)

Feature Name	Data Source	Construction Method	Operational Definition / Range
txt_unpainted	Text Description	Regex / Rule-Based	Binary (1 = mentions unpainted; 0 = otherwise)
txt_title_deeds	Text Description	Regex / Rule-Based	Binary (1 = mentions title deed or cessions; 0 = otherwise)
tfidf_dim_1 ... 30	Text Description	TF-IDF (30 n -grams)	Continuous TF-IDF term weights for top supervised description phrases
img_vis_painted	Images (./images/)	CLIP Zero-Shot / Vision	Continuous Probability Score [0.0,1.0] of painted interior/exterior walls
img_vis_tiled	Images (./images/)	CLIP Zero-Shot / Vision	Continuous Probability Score [0.0,1.0] of tiled flooring
img_vis_ceiling	Images (./images/)	CLIP Zero-Shot / Vision	Continuous Probability Score [0.0,1.0] of installed ceiling
img_vis_kitchen	Images (./images/)	CLIP Zero-Shot / Vision	Continuous Probability Score [0.0,1.0] of completed/fitted kitchen
img_vis_paved	Images (./images/)	CLIP Zero-Shot / Vision	Continuous Probability Score [0.0,1.0] of paved driveway/yard
img_vis_completion	Images (./images/)	CLIP Zero-Shot / Vision	Continuous Probability Score [0.0,1.0] of complete structural state